

- There are multiple level user to database
 - ⌘ **User ---- DNS --- Web --- Application --- DB**
 - ⌘ At user's browser level, caching happens. Means browser store the DNS information so that next time it'll not have to request for resolving the domain.
 - ⌘ At **Web** layer, **CDN** remains. Lets assume you want to serve some static files keeps in S3 bucket.
 - ⌘ So, first when you get the file, store it in a place and serve those files when required without going to the S3 bucket.
 - ⌘ This is done with **CDN**. (Content Delivery Network)
 - ⌘ At **Application** layer, you can use **Redis**, **Memcached** or some other caching tools so that you'll not have to make DB calls multiple times to fetch the same data.
 - ⌘ Even at **DB** layer you can do caching.
- **DNS**
 - ⌘ When a user requests a website, their system or browser queries DNS servers to translate the domain name into an IP address.
 - ⌘ Caching these results means that repeated requests for the same domain are resolved faster, reducing the need for repetitive network looks up and improving overall browsing efficiency.
 - ⌘ This is particularly beneficial for reducing latency and enhancing user experience by avoiding delays in domain resolution.
- **Web**
 - ⌘ Accelerate retrieval of web content from web/app servers.
 - ⌘ Web caching enhances performance by storing copies of frequently accessed web pages, images, and other content.
 - ⌘ When users request content, the cache delivers it quickly without needing to retrieve it from the origin server, which reduces load times and server strain.
 - ⌘ This not only speeds up content delivery but also reduces bandwidth usage and server costs.
 - ⌘ This is typically implemented using CDN.
 - ⌘ Web caching is crucial for improving the responsiveness of web applications and ensuring a smoother, faster user experience.
 - ⌘ **What it caches? : Static files (images, CSS, JS), HTML pages, Public API responses**
 - ⌘
 - ⌘

➤ **Application** caching (Redis, Memcached, ..etc)

- ⌘ What it caches? : DB query results, User sessions, Computed objects, Aggregations, Authorization info

➤ Simple difference between **CDN** and **Redis/Memcached/** (other application service) is:

- ⌘ **CDN** caches the things between **client** and **server**
- ⌘ **Redis / Memcached** cache data **inside your backend**, before hitting the database.

⌘ **Browser ⇔ CDN ⇔ App Server ⇔ Redis ⇔ Database**

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➤ **Cache terminologies**

- ⌘ Cache HIT and Cache MISS
- ⌘ Cache eviction policies: LRU, LFU, FIFO, TTL bases
- ⌘ Cache expiry and TTL
- ⌘ Cache loading strategies: Lazy loading, Eager loading or Cache warming
- ⌘ Cache size and capacity

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Caching in Spring Boot

➤ There are 2 annotations.

⌘ **@EnableCaching**

- ⌘ It's a **class** level annotation.
- ⌘ It should be written in a component.
- ⌘ Avoid writing this in the main method's application class, otherwise it'll make **caching mandatory**.
- ⌘ By default it setup a **CacheManager** and creates *in-memory* cache using one *concurrent Hashmap*.

⌘ **@Cacheable**

- ⌘ It's a **method** level annotation.
- ⌘ Its used with a method whose response has to be cached.
- ⌘ Spring boot manages the request and response of the method to the cache that is specified in the annotation attribute.
- ⌘ **It just see the cache, if the data present then just return that.**

⌘ **@CachePut**

- ⌘ It is also a **method** level annotation.
- ⌘ It is used to update the cache **after invoking the method**.
- ⌘ By doing this, the result is put in the cache and the method is executed.
- ⌘ It has same attributes of **@Cacheable** annotation.

⌘ **@CacheEvict**

- ⌘ It is also a **method** level annotation.
- ⌘ It is used to **remove the data from the cache**.
- ⌘ When the method is annotated with this annotation, then the method is executed and the cache will be removed/evicted.

➤ Example

```
@Component
public class MyMathService {

    @Cacheable("piDecimals")
    public int computePiDecimal(int precision) {
        ...
    }
}
```

- Cache is just like a **HashMap** which stores in *key-value* pair.

- ↗ Normally redis stores data like this

```
users::10 → {"id":10,"name":"Alok","email":"alok@example.com"}
```

- ↗ Lets assume the method is

```
@Cacheable(value = "users", key = "#id")
public User getUser(Long id) {
    return new User(10, "Alok", "alok@example.com");
}
```

- ↗ Here, the “value” field inside @Cacheable (**not the value inside redis**) represents the method (getUser)
- ↗ and “key” field inside @Cacheable represents the argument.
- ↗ These both combined makes the **key of Redis**, and the **value of Redis** against this key will be like

```
{"id":10,"name":"Alok","email":"alok@example.com"}
```

- ↗ So, “**value**” field inside @Cacheable is not the *value* of Redis cache.
- ↗ Instead of value, you can write cacheNames as well. Both are aliases of each other.

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➤ Implementing Redis caching in spring boot application

- First create a config file and annotate it with `@EnableCaching`, because you shouldn't write this annotation in the main method's class.

```
@Configuration
@EnableCaching
public class CacheConfig {
}
```

- Annotate the cacheable method with `@Cacheable`

```
@Cacheable(value = "employee", key = "#id")
public EmployeeDto getEmployeeById(Long id) {
```

- ⚡ Note that #, without this it'll not take the value of the argument.
- ⚡ Here, Redis key will be like: **employee::3** (if id = 3)
- ⚡ If you want to be more specific, like if the method has more arguments and you want to be specific with more arguments then

```
@Cacheable(value = "employee", key = "{#id, #name}")
public EmployeeDto getEmployeeById(Long id, String name) {
```

it'll create keys like: **employee::[5, alok]** (id = 5, name = "alok")

- ⚡ Its SePL query, **{ }** means list.
- ⚡ It'll become a **composite key**. (id, name) (Object composite)
- ⚡ **It is safer to use.**
- ⚡ Also, another method of writing this is

```
@Cacheable(value = "employee", key = "#id ':' #name}")
public EmployeeDto getEmployeeById(Long id, String name) {
```

it'll create keys like: **employee::5:alok**

- ⚡ Its also **composite key** (String composite)
- ⚡ If you have a lot of filters (to be passed inside *key* field) then you can use a custom key generator function and pass like

```
@Cacheable(value = "employee", keyGenerator = "customKeyGen")
public EmployeeDto getEmployeeById(Long id, String name) {
```

- ⚡ It sends all the arguments to that **customKeyGen** method.
- ⚡ One more method is *hashing*, you can create a token and store that as the **key in redis cache**.
- ⚡ Either you can use custom key generator method where you'll return the hash.

- Or you can use SePL language directly

```
@Cacheable(value = "search", key = "T(com.example.HashUtil).sha256(#request.toString())")
```

- Annotate with `@CachePut` annotation

```
@CachePut(cacheNames = "employee", key = "#result.id")  
public EmployeeDto createNewEmployee( @NotNull EmployeeDto employeeDto)
```

```
@CachePut(cacheNames = "employee", key = "#id")  
public EmployeeDto updateEmployee(Long id, @NotNull EmployeeDto employeeDto)
```

- ⌘ `#result.id` because the return type of `createNewEmployee` method is `EmployeeDto`,

```
@Data 24 usages  Alok Ranjan Joshi  
public class EmployeeDto implements Serializable {  
    private Long id;  
    private String email;  
    private String name;  
    private Long salary;  
}
```

even if it is private and the getter method is `getId()`, so instead of `#result.id` we should write `#result.getId()`, but NO.

Internally SePL invokes the getter method like `#result.getId()` when we write `#result.id`

- ⌘ **Now, whenever we'll create a new employee or update an employee, it'll write the latest data into the cache after the method is executed.**
- ⌘ **NOTE: it'll store the data in Cache only if the method is executed successfully, in case of any exception it doesn't store the data in cache.**

- Annotate with `@CacheEvict` annotation

```
@CacheEvict(cacheNames = "employee", key = "#id")  
public void deleteEmployee(Long id) {
```

- ⌘ It'll delete the data from the cache if the method `deleteEmployee` is executed successfully.



Internal working of Caching in Spring Boot

- There is an interface **CacheManager** which has to be implemented by the cache provider, in our case it is **Redis**

```
public interface CacheManager { 3 usages
    @Nullable 6 implementations
    Cache getCache(String name);

    Collection<String> getCacheNames();
}
```

```
public class RedisCacheManager extends AbstractTransactionSupportingCacheManager {
    protected static final boolean DEFAULT_ALLOW_RUNTIME_CACHE_CREATION = true;
    private final boolean allowRuntimeCacheCreation;
    private final RedisCacheConfiguration defaultCacheConfiguration;
    private final RedisCacheWriter cacheWriter;
    private final Map<String, RedisCacheConfiguration> initialCacheConfiguration;
```

(**AbstractTransactionSupportingCacheManager** extends **CacheManager**)

- **defaultCacheConfiguration** : it is to set different types of configs like prefix, expiry, reset the expiry if the cache hits again ..etc

```
public class RedisCacheConfiguration { no usages
    protected static final boolean DEFAULT_CACHE_NULL_VALUES = true;
    protected static final boolean DEFAULT_ENABLE_TIME_TO_IDLE_EXPIRATION = false;
    protected static final boolean DEFAULT_USE_PREFIX = true;
    protected static final boolean DO_NOT_CACHE_NULL_VALUES = false;
    protected static final boolean DO_NOT_USE_PREFIX = false;
    protected static final boolean USE_TIME_TO_IDLE_EXPIRATION = true;
    private final boolean cacheNullValues;
    private final boolean enableTimeToIdle;
    private final boolean usePrefix;
    private final CacheKeyPrefix keyPrefix;
    private final ConversionService conversionService;
    private final RedisSerializationContext.SerializationPair<String> keySerializationPair;
    private final RedisSerializationContext.SerializationPair<Object> valueSerializationPair;
    private final RedisCacheWriter.TtlFunction ttlFunction;
```

- **initialCacheConfiguration** : it is to set the default configurations for different caches, like for “employee” one type of configuration, for “department” another type of configuration

```
Map<String, RedisCacheConfiguration> configs = new HashMap<>()

configs.put("employee",
    RedisCacheConfiguration.defaultCacheConfig()
        .entryTtl(Duration.ofMinutes(10)));

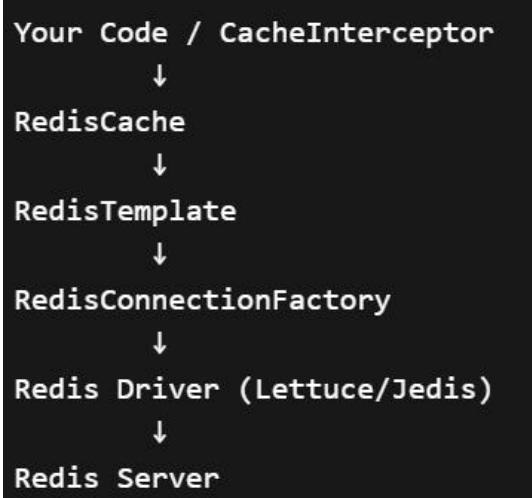
configs.put("users",
    RedisCacheConfiguration.defaultCacheConfig()
        .entryTtl(Duration.ofHours(1)));
```


- This **CacheManager**'s bean is auto-configured by spring, if you want custom configuration then you can create the bean of **RedisCacheManager**.
- This is how you can create custom RedisCacheManager bean

```
@Bean new *
public CacheManager cacheManager(RedisConnectionFactory redisConnectionFactory) {
    RedisCacheConfiguration redisCacheConfiguration = RedisCacheConfiguration.defaultCacheConfig()
        .prefixCacheNameWith( prefix: "my-test-")
        .entryTtl( ttl: Duration.ofSeconds( seconds: 60))
        .enableTimeToIdle()
        .serializeKeysWith(
            keySerializationPair: RedisSerializationContext
                .SerializationPair
                .fromSerializer( serializer: new StringRedisSerializer())
        )
        .serializeValuesWith(
            valueSerializationPair: RedisSerializationContext
                .SerializationPair
                .fromSerializer( serializer: new GenericJackson2JsonRedisSerializer())
        );

    return RedisCacheManager.builder(redisConnectionFactory)
        .cacheDefaults( defaultCacheConfiguration: redisCacheConfiguration)
        .build();
}
```

- ⌘ **RedisConnectionFactory** (implemented by **LettuceConnectionFactory**) creates and manages connections that Spring uses to talk to Redis.
 - ⌘ Without this spring cannot execute GET, SET, DEL, TTL, ..etc.
- ⌘ **RedisCacheConfiguration** is the one which are (mentioned in the previous page), it adds the default configurations for the redis cache manager.
- ⌘ **serialize** are for the type in which keys and values will be stored in redis db.



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- Internal Flow
- CacheAutoConfiguration (auto-configuration class)
 - ⌘ CacheManagerCustomizers, CacheManagerValidator,
 - ⌘ CacheManagerCustomizers contains list of CacheManagerCustomizer
 - ⌘ CacheManagerCustomizer takes generic (T extends CacheManager), it contains a manager *customize* which takes CacheManager or one of its sub classes as its argument.
 - ⌘ So, if we create CacheManager bean by our own, then it'll take that CacheManager bean instead of the default one.
 - ⌘ Now the question is, we create only one bean of CacheManager, then how it's a list

```
public class CacheManagerCustomizers { no usages
    private final List<CacheManagerCustomizer<?>> customizers;
```

- ⌘ So, this is not list of CacheManager itself, its a list of CacheManagerCustomizer

```
@FunctionalInterface no usages
public interface CacheManagerCustomizer<T extends CacheManager> {
    void customize(T cacheManager);
}
```

- ⌘ We can have multiple CacheManagerCustomizer

```
@Bean
CacheManagerCustomizer<RedisCacheManager> ttlCustomizer() {
    return cm -> cm.setDefaultExpiration(Duration.ofMinutes(5));
}

@Bean
CacheManagerCustomizer<RedisCacheManager> metricsCustomizer() {
    return cm -> enableMetrics(cm);
}
```

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- Spring Boot is built on this philosophy:

“The framework owns object creation,
developers only tweak behavior.”

Spring scans beans



Finds CacheManagerCustomizer bean



CacheAutoConfiguration collects it via ObjectProvider



Adds it to CacheManagerCustomizers list



When CacheManager is created



Boot calls customizer.customize(cacheManager)

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- Customizer interface is to customize a already created object with some different parameters. Consider the below example

```
public class Server {

    private int port = 8080;
    private int timeout = 30;
    private boolean compressionEnabled = false;
    private String environment = "dev";
    private boolean metricsEnabled = false;

    // setters
    public void setPort(int port) { this.port = port; }
    public void setTimeout(int timeout) { this.timeout = timeout; }
    public void setCompressionEnabled(boolean compressionEnabled) {
        this.compressionEnabled = compressionEnabled;
    }
    public void setEnvironment(String environment) { this.environment = environment; }
    public void setMetricsEnabled(boolean metricsEnabled) { this.metricsEnabled = metricsEnabled; }

    @Override
    public String toString() {
        return "Server{" +
            "port=" + port +
            ", timeout=" + timeout +
            ", compressionEnabled=" + compressionEnabled +
            ", environment='" + environment + '\'' +
            ", metricsEnabled=" + metricsEnabled +
            '}';
    }
}
```

```
@FunctionalInterface
public interface ServerCustomizer {
    void customize(Server server);
}
```

```
public class PortCustomizer implements ServerCustomizer {
    @Override
    public void customize(Server server) {
        server.setPort(9090);
    }
}
```

```
public class TimeoutCustomizer implements ServerCustomizer {
    @Override
    public void customize(Server server) {
        server.setTimeout(60);
    }
}
```

```
public class CompressionCustomizer implements ServerCustomizer {
    @Override
    public void customize(Server server) {
        server.setCompressionEnabled(true);
    }
}
```

```
public class MetricsCustomizer implements ServerCustomizer {
    @Override
    public void customize(Server server) {
        server.setMetricsEnabled(true);
    }
}
```

```
public class EnvironmentCustomizer implements ServerCustomizer {
    @Override
    public void customize(Server server) {
        server.setEnvironment("production");
    }
}
```

- Customizer takes the actual object that has to be customized, and customize its params so that you need not change the properties of the object manually.

```
import java.util.List;

public class ServerFactory {

    public static Server createServer(List<ServerCustomizer> customizers) {

        // Framework creates object with defaults
        Server server = new Server();

        // Apply all customizers
        for (ServerCustomizer customizer : customizers) {
            customizer.customize(server);
        }

        return server;
    }
}
```

- Server factory takes list of customizers (here ServerCustomizer; because it is the customizer interface of the Server class) and create an object and change its values according to those customizers.
- You can have multiple customizers for multiple properties of an object.

```
public static void main(String[] args) {

    List<ServerCustomizer> customizers = List.of(
        new PortCustomizer(),
        new TimeoutCustomizer(),
        new CompressionCustomizer(),
        new EnvironmentCustomizer(),
        new MetricsCustomizer()
    );

    Server server = ServerFactory.createServer(customizers);

    System.out.println(server);
}
```

- Now, you'll take all the customizers (that you need) in a list and create the object and call the *customize* method of all those customizers to configure your created object.
- HENCE: Customizer doesn't help in creation of an object (bean in spring boot), rather they changes the properties of that object after the object being created.

- In spring security, when we have to configure the *formLogin* part, we can directly pass **Customizer.withDefaults()** which by default set the properties

```
http.formLogin(Customizer.withDefaults());
```

instead we had to write like the below

```
http.formLogin(form -> {});
```

```
@Bean
SecurityFilterChain security(HttpSecurity http) throws Exception {
    http
        .authorizeHttpRequests(auth -> auth.anyRequest().authenticated())
        .formLogin(Customizer.withDefaults())
        .httpBasic(Customizer.withDefaults());

    return http.build();
}
```

-
- ⌘ This **Customizer.withDefaults()** also returns a Customizer object.

```
public interface Customizer<T> {

    void customize(T t);

    static <T> Customizer<T> withDefaults() {
        return t -> { };    // no changes
    }
}
```

- ⌘ formLogin() takes customizer and customize the created object according to that.
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